

George Bassem

Backend AI Engineer — LLM Systems, Neural Networks & AI Agents

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SUMMARY

AI engineer and Computer Science undergraduate focused on the backend systems that make LLMs and neural networks run in production: distributed training, inference serving, and the data/RAG pipelines that feed them. I build AI agents, fine-tune and deploy LLMs across multi-GPU hardware, and design retrieval-augmented systems grounded in external knowledge — working from first principles on transformers, attention optimization, FLOP accounting, and 40 GB+ dataset pipelines.

PROJECTS

Facial Emotion Recognition — Distributed Training Pipeline (1st Place, Kaggle)

- Shipped a **multi-GPU distributed training pipeline** that fine-tuned **ViT and SigLIP2** vision transformer backbones, taking **1st place** in a private Kaggle emotion-detection competition.
- Scaled training across **2 GPUs with PyTorch DDP** via **torchrun**, tracking down and fixing NCCL ALLREDUCE timeouts caused by rank-asymmetric tensor sizes and a distributed-evaluation bug that was silently halving reported accuracy.
- Built the backend data layer end to end — **WeightedRandomSampler, DistributedSampler**, parallel-hash deduplication, and conversion to **HuggingFace DatasetDict** format — then replaced the HF Trainer with a **native PyTorch training loop** for tighter control over scheduling and loss accumulation.

Mini GPT — Transformer Implemented from Scratch

- Built a GPT-style transformer's full backend from the ground up — **tokenizer, multi-head self-attention, and training loop** — to master how LLM internals actually execute, including FLOP accounting and gradient flow through attention.

Retrieval-Augmented Generation (RAG) Service

- Designed and shipped a **RAG backend** pairing a **vector retrieval layer** with an LLM to return grounded, source-cited answers instead of unverified generations.

Multi-GPU LLM Inference Deployment

- Deployed LLMs locally across **multi-GPU hardware**, applying **FlashAttention-3** and memory optimizations to cut inference latency and serving cost.

LLM Fine-Tuning for Custom Tasks

- Adapted pre-trained LLMs to custom tasks and datasets using **fine-tuning** and **chain-of-thought prompting**, measurably lifting task-specific performance.

Autonomous AI Agents

- Built autonomous agents that combine **LLM reasoning with tool use** to plan and execute multi-step backend workflows without manual intervention.

CORE STACK

Python (PyTorch, TensorFlow) · SQL · PHP (Laravel) · Multi-GPU distributed training (DDP, torchrun, NCCL) · FlashAttention-3 & attention optimization · RAG & vector retrieval · LLM fine-tuning · HuggingFace ecosystem · FLOP & convex optimization analysis · Large-scale data pipelines (40 GB+)

HONORS & AWARDS

- 1st Place — Face Emotion Detection, Kaggle private competition
- Best Deep Learning Member — ApplAI (2024)
- Honor Certificate — El Wayly Educational Administration

EDUCATION

Ain Shams University

Expected May 2027

B.Sc. in Computer Science — Artificial Intelligence · CGPA 3.70 / 4.0

CERTIFICATIONS

- Machine Learning Specialization — Coursera (DeepLearning.AI & Stanford)